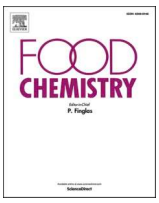
Food Chemistry 424 (2023) 136429



Contents lists available at ScienceDirect

Food Chemistry

journal homepage: www.elsevier.com/locate/foodchem





Characterization of the differences in aroma-active compounds in strong-flavor Baijiu induced by bioaugmented *Daqu* using metabolomics and sensomics approaches

Yu Mu [a], Jun Huang [a], Rongqing Zhou [a,b,*], Suyi Zhang [b,c], Hui Qin [b,c], Hanlan Tang [b,c], Qianglin Pan [a], Huifang Tang [a]

[a] College of Biomass Science and Engineering, Sichuan University, Chengdu 610065, China
[b] National Engineering Research Center of Solid-State Manufacturing, Luzhou 646000, China
[c] Luzhou Lao Jiao Co., Ltd., Luzhou 646699, China

ARTICLE INFO

Keywords:
Bioaugmented *Daqu*
Strong-flavor Baijiu
GC×GC-TOF/MS
GC–O/MS
Multivariate statistical analysis

ABSTRACT

Bioaugmenting *Daqu* is an effective strategy to improve the quality of Baijiu, but its effect on overall flavor profiles and aroma-active compounds is unknown. Here, 168 volatiles were determined from fresh strong-flavor Baijiu (SFB) and bioaugmented *Daqu* increased their diversity and altered the flavor characteristics. Among 49 odorants identified by aroma extraction dilution analysis, 29 aroma-active compounds had odor activity values ≥1, of which 18, 8, and 3 components exhibited the highest content in the SFB fermented by fortified-, space- and conventional-*Daqu*, respectively. The contribution of increasing ethyl hexanoate and decreasing ethyl lactate of fresh SFB by bioaugmented *Daqu* was confirmed, and their content changed from 4650 and 1890 mg/L (conventional-SFB) to 6680 and 1760 mg/L (fortified-SFB) and 6130 and 1710 mg/L (space-SFB). Meanwhile, the discriminators among different samples were determined by multivariate statistical analysis. These findings are beneficial for the optimization and improvement of Baijiu aroma.

1. Introduction

Strong-flavor Baijiu (SFB) with unique flavor attributes such as pit flavor, purity, coordination and long aftertaste account for over 70% of the Baijiu market (Wang, Chen, Wu, & Zhao, 2022). Its flavor compounds mainly originated from fermented grain (*Zaopei*), a niche of community succession and metabolite synthesis, initiated and regulated by *Daqu* microbiota (Zou, Zhao, & Luo, 2018). Hence, bioaugmented *Daqu* based on inoculating functional strains or microbiota had been used to improve the quality and flavor of fresh Baijiu. For example, *Daqu* bioaugmented by synthetic microbiota consisting of *Bacillus velezensis* and *Bacillus subtilis* significantly increased the content of esters and aromatics in fresh SFB (He, Huang, Zhou, Wu, & Jin, 2019). Similarly, *Xiaoqu* inoculating *Monascus purpureus* not only increased the content of total ester but also decreased that of higher alcohols in the fresh Baijiu, thus improving the flavor balance (Zhu et al., 2022). However, high-resolution techniques combined with sensory analysis are required to fully understand the contribution of bioaugmented *Daqu* to the flavor profiles and aroma-active compounds of fresh Baijiu.

The relationship between volatiles and sensory perception has been a hot topic in food flavor chemistry (Lemarcq, Van de Walle, Monterde, Sioriki, & Dewettinck, 2021). Despite the identification of over 2,000 volatiles in Baijiu, it remains challenging to accurately assess their contribution to sensory. Compared with gas chromatographic–mass spectrometry (GC–MS) commonly used for evaluating the volatile compounds of Baijiu, GC × GC greatly improves the resolution capacity of the peak through the two-dimensional expansion of polar and non-polar columns, and coupling with time-of-flight mass spectrometry (TOF/MS) enhances sensitivity, acquisition speed and mass range (Amaral & Marriott, 2019). For instance, He et al. (2020) determined the regional features of flavor constituents in SFB between Sichuan and Jianghuai based on 262 volatiles identified by GC × GC-TOF/MS. Additionally, the aging markers and the differences in flavor profiles of Qingke Baijiu with different ages were also revealed by the combination of GC × GC-TOF/MS and metabolomics approach (Wang, Song et al., 2021). Gas chromatography–olfactory apparatus (GC–O) is widely used for identifying the compounds that directly contribute to the aroma of Baijiu due to its high objectivity (Jia et al., 2020). Hong et al. (2021)

* Corresponding author at: College of Biomass Science and Engineering, Sichuan University, Chengdu 610065, China.
E-mail address: zhourqing@scu.edu.cn (R. Zhou).

https://doi.org/10.1016/j.foodchem.2023.136429
Received 13 February 2023; Received in revised form 19 April 2023; Accepted 18 May 2023
Available online 22 May 2023
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using sensomics based on GC–O/MS and aroma extraction dilution analysis (AEDA) found that the differences in aroma features of SFB from different regions were caused by a few compounds with high odor activity values (OAVs). The sensomics also identified the aroma-active compounds and the odorants responsible for roasted and mud-like aromas in SFB (Dong et al., 2019; Zhao et al., 2018). These works laid methodology foundations for unraveling the differences in aroma characteristics of fresh SFB produced by *Daqu* with disparate attributes.

This study aimed to (1) characterize the effects of bioaugmented *Daqu* on the volatiles of fresh SFB using GC × GC-TOF/MS; (2) identify and assess the intensity of odorants through GC–O/MS combined with AEDA; (3) quantify and verify aroma-active compounds with polyphasic detection methods and OAVs; and (4) determine significant differential compounds and discriminators using multivariate statistical analysis. These results provide a methodology and theoretical foundation for enhancing the quality of traditional fermented food using the bioaugmentation technique.

2. Materials and methods

2.1. Baijiu fermentation and sampling

Three fresh Baijiu samples were produced in the Huangyi base of Luzhou Laojiao Co., Ltd. (Luzhou, Sichuan, China). Briefly, *Daqu* with different attributes were added to the steamed and cooled grains (sorghum) and mixed well, then the mixture was load into the mud pit (an underground vessel) for anaerobic fermentation. After 80 days of fermentation (April to June 2022), the second stage of fresh Baijiu, called "Erduan Jijiu", with high ethanol content (ca. 70% vol/vol) and flavor quality was collected during the distillation process of *Zaopei*. Before that, the *Zaopei* had been fermented in three rounds using corresponding *Daqu* in the identical mud pit to avoid the influence of *Huizao* (*Zaopei* from the previous round) and mud pit on the flavor profile of fresh Baijiu (Wang et al., 2022). The target samples using fortified-*Daqu* (F_*Daqu*) and space-*Daqu* (S_*Daqu*) were abbreviated as FB and SB, respectively, while the control sample using conventional-*Daqu* (C_*Daqu*) was called CB. Samples were stored at 4 °C until analysis. The differences in these *Daqu* were as follows: F_*Daqu* with the highest saccharification and fermentation powers was characterized by *Enterococcus*, *Bacillus* and *Thermomyces*; S_*Daqu* with the highest liquefaction and esterification powers was dominated by *Kroppenstedtia*, *Bacillus* and *Pichia*; while C_*Daqu* with the lowest enzyme activities was mainly composed of *Lactobacillus*, *Weissella* and *Aspergillus* (Mu et al., 2022a).

2.2. Reagents and chemicals

All chemical standards used in this study were GC grade or analytical reagent (AR) grade with at least 97% purity. Amyl acetate (internal standard, IS1), ethyl hexanoate, ethyl lactate, ethyl butanoate, ethyl pentanoate, propyl hexanoate, butyl hexanoate, isopentyl hexanoate, pentyl hexanoate, hexyl hexanoate, ethyl 2-methylpropanoate, ethyl 2-methylbutanoate, ethyl 3-methylbutanoate, ethyl 4-methylpentanoate, hexyl acetate, ethyl heptanoate, ethyl 2-hydroxy-4-methylvalerate, butanoic acid, hexanoic acid, 1-butanol, 1-hexanol and phenethyl alcohol were purchased from Aladdin Biochemical Technology Co., Ltd. (Shanghai, China). 2-Ethylbutyric acid (IS2), isoamyl acetate, isoamyl octanoate, ethyl nonanoate, ethyl benzoate, diethyl butanedioate, hexyl octanoate, ethyl 3-phenylpropanoate, acetic acid, 3-methyl-1-butanal, dimethyl trisulfide, 1-octanol and 4-methylphenol were purchased from Macklin Biochemical Technology Co., Ltd. (Shanghai, China). Ethyl octanoate, ethyl decanoate, ethyl hexadecanoate and 2-methyl-1-propanol were purchased from J&K Scientific Co., Ltd. (Beijing, China). 3-Methyl-1-butanol, benzaldehyde and C8-C40 *n*-alkanes were purchased from Sigma-Aldrich (Shanghai, China), while 2-octanol (IS3), ethyl acetate, ethyl *trans*-4-decenoate and ethyl phenylacetate were purchased from Dr.Ehrenstorfer GmbH (Augsburg, Germany), Fluka (Seelze, Germany), Alfa Aesar (Shanghai, China) and Tokyo Chemical Industry Co., Ltd. (Tokyo, Japan), respectively. Ethanol (HPLC grade), sodium chloride (AR) and anhydrous sodium sulfate (AR) were purchased from Sinopharm Chemical Reagent Co., Ltd. (Beijing, China).

2.3. Analysis of volatile compounds using GC × GC-TOF/MS

The volatile compounds in fresh Baijiu were extracted using headspace solid-phase microextraction (HS-SPME) following a modified method based on a previous study (Wang, Gao, Liu, Chen, & Xu, 2020). Briefly, fresh Baijiu was diluted with saturated NaCl solution to 14% ethanol by volume, 5 mL diluted sample was placed in a 20 mL headspace bottle and equilibrated for 15 min at 50 °C. The volatiles was then extracted using a 50/30 μm DVB/CAR/PDMS fiber (Supelco, Bellefonte, PA, USA) at 50 °C for 45 min while stirring at 400 r/min. The SPME fibers were desorbed at 250 °C for 5 min, and the volatile analysis was carried out using a GC × GC-TOF/MS system comprising an Agilent 7890A (Agilent Technologies, Santa Clara, CA, USA) and a Pegasus 4D time of flight mass spectrometer (LECO Corp., St. Joseph, MI, USA).

The system used for volatile compound analysis consisted of a primary DB-WAX polar column (30 m × 0.25 mm × 0.25 μm; Agilent Technologies, Santa Clara, CA, USA) and a secondary DB-17Si MS semi-polar column (1.8 m × 0.18 mm × 0.18 μm; Agilent Technologies, Santa Clara, CA, USA) using high-purity helium (>99.999%) as the carrier gas at 1.00 mL/min. The temperature program for the primary column was as follows: held at 45 °C for 3 min, ramped at 5 °C/min to 230 °C and held for 10 min. The secondary column was maintained at 15 °C higher than the primary column, with the modulator offset by + 20 °C relative to the primary oven. A modulation period of 5.0 s (hot pulse of 1.0 s) was used with cryogenic trap cooled to − 196 °C by liquid nitrogen. The TOF/MS parameters included ion source of 230 °C, transmission line of 240 °C, ionization mode of EI (70 eV), voltage of 1500 V, mass range of 35–400 amu and acquisition rate of 100 spectra/s.

Data processing was carried out using ChromaTOF software (v 4.32; LECO Corp., St. Joseph, MI, USA). First, automated peak finding and spectral deconvolution were performed, with a signal to noise (S/N) ratio threshold set at 50. Then, compounds with a similarity index (SI) and reverse similarity index (RSI) >800 were retained by comparing them to reference databases (NIST 2014 and Wiley 9), while column bleeding and compounds detected only once in triplicate were removed. Finally, the first dimension linear retention indices (LRIs) of each compound were determined using *n*-alkanes, and compounds with a maximum deviation within 50 were considered reliable based on comparison with the literature. The relative content of each compound was determined using peak area normalization.

2.4. Identification of aroma compounds by sensomics approach

2.4.1. GC–O/MS analysis

The aroma compounds were extracted using HS-SPME, as described in Section 2.3. To identify the aroma compounds, a GC–MS system (QP2010 SE; Shimadzu Co., Kyoto, Japan) equipped with a DB-WAX column and an olfactory detection port (ODP3; Gerstel, Mülheim an der Ruhr, Germany) was used. The temperatures of the GC injector and oven were the same as those used in the first-dimensional GC × GC. The MS was operated in an EI mode (70 eV) and the mass range of detection was set at 35–400 amu with 1 scan/s. The temperatures of the ion source and transmission line were set at 230 °C and 240 °C, respectively.

The effluent from the capillary column was mixed with warm, humid air for sniffing at a 1:1 ratio and introduced into the olfactory detection port (ODP) with a temperature of 230 °C. Three trained panelists (two male and one female) sniffed over 40 aroma-active compounds at concentrations 10 times their odor threshold using GC–O to identify the aroma compounds. During GC operation, the panelists placed their noses near the ODP to obtain retention times and descriptions for each aroma compound. Each sample was analyzed three times by each panelist, and

only compounds with a frequency ≥ 2 were considered for further analysis. The retention indices (RIs) were calculated using *n*-alkanes under the same conditions. The aroma compounds were identified by comparing with the NIST library, aromas, and RI on the DB-WAX column with those of the authentic standards.

2.4.2. Aroma extract dilution analysis (AEDA)

To determine the intensity of the aroma compounds, we used AEDA as described by Zhu et al. (2020). Volatile compounds extracted by HS-SPME were stepwise diluted using different split ratios (1:1, 3:1, 9:1, 27:1, 81:1, and 243:1), and then analyzed using GC–O/MS under the same conditions. The flavor dilution factor (FD) of each odorant was defined as the highest split ratio at which the odorant could be perceived at the sniffing port. We assigned an FD value to each odorant as the highest split ratio at which it could be perceived using GC–O, and the maximum FD value observed was 243.

2.5. Quantitative analysis of aroma compounds

2.5.1. GC with flame ionization detector (FID)

Seven different compounds with concentrations exceeding 100 mg/L were determined by direct injection (DI) combined with GC–FID, which equipped a DB-WAX column. To more accurately replicate the conditions of the authentic samples, a series of mixed standard working solutions with varying concentrations were created using a 70% ethanol solution by volume. The highest and lowest concentrations measured for each of the seven compounds were as follows: ethyl hexanoate (9000/450 mg/L), ethyl acetate (6000/300 mg/L), ethyl lactate (4000/200 mg/L), ethyl butyrate (900/45.0 mg/L), hexanoic acid (1000/50.0 mg/L), acetic acid (1000/50.0 mg/L) and butanoic acid (600/30.0 mg/L). Before analysis, 0.2 g of anhydrous sodium sulfate was added to 1.0 mL of fresh Baijiu samples or the standard working solutions, and two internal standards (IS1 and IS2) were added to achieve final concentrations of 400 and 80.0 mg/L, respectively. After standing at 4 °C for 6 h, samples were filtered by a 0.45 μm nylon membrane and 1.0 μL was analyzed by GC–FID with split mode (20:1). High-purity nitrogen (>99.999%) was used as the carrier gas at a constant flow rate of 1.0 mL/min. The oven temperature was programmed to 40 °C for 4 min, ramped at 20 °C/min to 100 °C and held for 10 min, and then increased to 200 °C at 40 °C/min and held for 3 min. The temperature of both the injector and detector were set at 250 °C, and the flow rates of hydrogen and air were 40 and 450 mL/min, respectively.

2.5.2. Direct injection combined with GC–MS/MS

The remaining aroma compounds were quantified using a Trace 1300-TSQ 9000 GC–MS/MS (Thermo Fisher Scientific, Waltham, MA, USA) equipped with a VF-WAX-MS column (30.0 m × 0.25 mm × 0.25 μm; Agilent Technologies, Santa Clara, CA, USA). Helium (>99.999%) was used as the carrier gas at a constant flow rate of 1.0 mL/min, and the injector temperature and split ratio were set at 250 °C and 10:1, respectively. The oven temperature was programmed to 40 °C for 5 min, ramped at 4 °C/min to 100 °C, and then increased to 230 °C at 6 °C/min and held for 10 min. The MS parameters included ion source of 250 °C, transmission line of 300 °C, ionization mode of EI (70 eV) and mass range of 35–400 amu (1 scan/s). The selective ion monitoring (SIM) mode with higher sensitivity and better precision was used to quantify aroma compounds (Du, Li et al., 2021). Target compounds were grouped into two categories based on semi-quantitative results, with maximum/minimum concentrations of the series standard working solution of group A and group B being 600/1.00 mg/L and 50.0/0.10 mg/L, respectively. Anhydrous sodium sulfate (0.2 g) and an appropriate amount of IS1-IS3 were added to 1.0 mL fresh Baijiu samples or working standard solutions to make a final concentration of 80.0 mg/L (group A) and 40.0 mg/L (group B), and the mixture was analyzed after standing and filtering.

2.6. Statistical analysis

The concentrations of aroma compounds were measured in triplicate and presented as mean ± standard deviation. The bar plot and Venn plot were generated using Origin 2019 (OriginLab Corp., Northampton, Massachusetts, USA). Principal component analysis (PCA) and orthogonal partial least squares discriminant analysis (OPLS-DA) were performed on the log-transformed quantitative data set by Smica 14.1 (Umetrics AB, Umea, Vasterbotten, Sweden). Linear discriminant analysis effect size (LEfSe) was performed using an open-access pipeline (https://huttenhower.sph.harvard.edu/galaxy/) with a threshold of LDA > 2.0 and $p < 0.05$ (Segata et al., 2011). One-way ANOVA with a significance level of $p < 0.05$ was employed to assess the differences between samples using SPSS 24.0 (IBM Corp., Armonk, NY, USA).

3. Results and discussion

3.1. Effects of Daqu attributes on the volatile profiles of fresh SFB

The effect of bioaugmented *Daqu* on the volatile profiles of fresh Baijiu was investigated using HS-SPME-GC × GC-TOF/MS. The two-dimensional total ion chromatogram revealed the distribution pattern of esters, aldehydes, alcohols, and acids as the number of carbons increases (Fig. 1), which avoided the coelution of multiple compounds on the one-dimensional time and obtained higher resolution for flavor profiling (Jia et al., 2020). Additionally, the relative content can be evaluated intuitively based on the area and color of chromatographic peaks in a two-dimensional chromatogram (Amaral & Marriott, 2019). More than 500 chromatographic peaks were detected in the samples, with 168 different compounds identified in 9 classes based on S/N > 50, MS similarity > 800, and RI deviation < 50. These compounds included 67 esters, 27 alcohols, 16 aromatics, 16 aldehydes, 14 acetals, 13 ketones, 9 acids, 3 furans and 3 sulfur-containing compounds (Table S1).

The relative content of ethyl hexanoate (C6) was the highest among the homologous ethyl esters of *C*2-C10 and C12, followed by ethyl acetate (*C*2) and ethyl butyrate (C4), which are considered the skeleton components of SFB (Zou et al., 2018). Other important esters included isoamyl acetate (1), butyl hexanoate (2), isoamyl hexanoate (3) and hexyl hexanoate (4), which impart fruity and sweet odors to SFB (Wang et al., 2022). Straight-chain aldehydes (*C*2-C12) and alcohols (C3-C9) were also detected, governed by acetaldehyde (*C*2) and 1-hexanol (C6), respectively, but their relative contents were lower than those of branched-chain aldehydes and alcohols, namely 3-methyl-1-butanal (5) and 3-methyl-1-butanol (6). These two branched-chain compounds are produced from the transamination and decarboxylation of the leucine and contribute to the nutty and malt aromas of SFB (Hazelwood, Daran, Maris, Pronk, & Dickinson, 2008; Wang et al., 2022). The homologous series of straight-chain monocarboxylic fatty acids from *C*2 and C4-C8 were identified, with hexanoic acid (C6) being an important aroma-active compound in SFB, contributing to sweaty and cheesy odors (Fan & Qian, 2006). Additionally, 1,1-diethoxy-3-methylbutane (7) with high relative content also contributed to the fruity odor of SFB (Zhao et al., 2018). Three sulfur-containing compounds were detected and dominated by dimethyl trisulfide (8), which has been regarded as the key aroma-active compound in sauce- and sesame-flavor Baijiu as its odor threshold (0.36 μg/L) is extremely low (Ma, Yi, & Zhu, 2020; Li et al., 2019).

As shown in Fig. 2A-B, the number and relative content of volatiles in fresh SFB were significantly affected by bioaugmented *Daqu*. The total number increased from 128 in CB to 135 and 147 in FB and SB, respectively, with esters (82.78–85.99%), alcohols (6.35–7.21%), aldehydes (2.54–3.36%) and acetals (1.20–4.96%) as the dominant. SB and FB had higher numbers of esters, alcohols, aromatics and ketones, while CB had more acetals. Similarly, a higher content of esters, alcohols and aromatics was observed in SB and FB, whereas acetals and aldehydes were enriched in CB. It is well known that the volatile components in

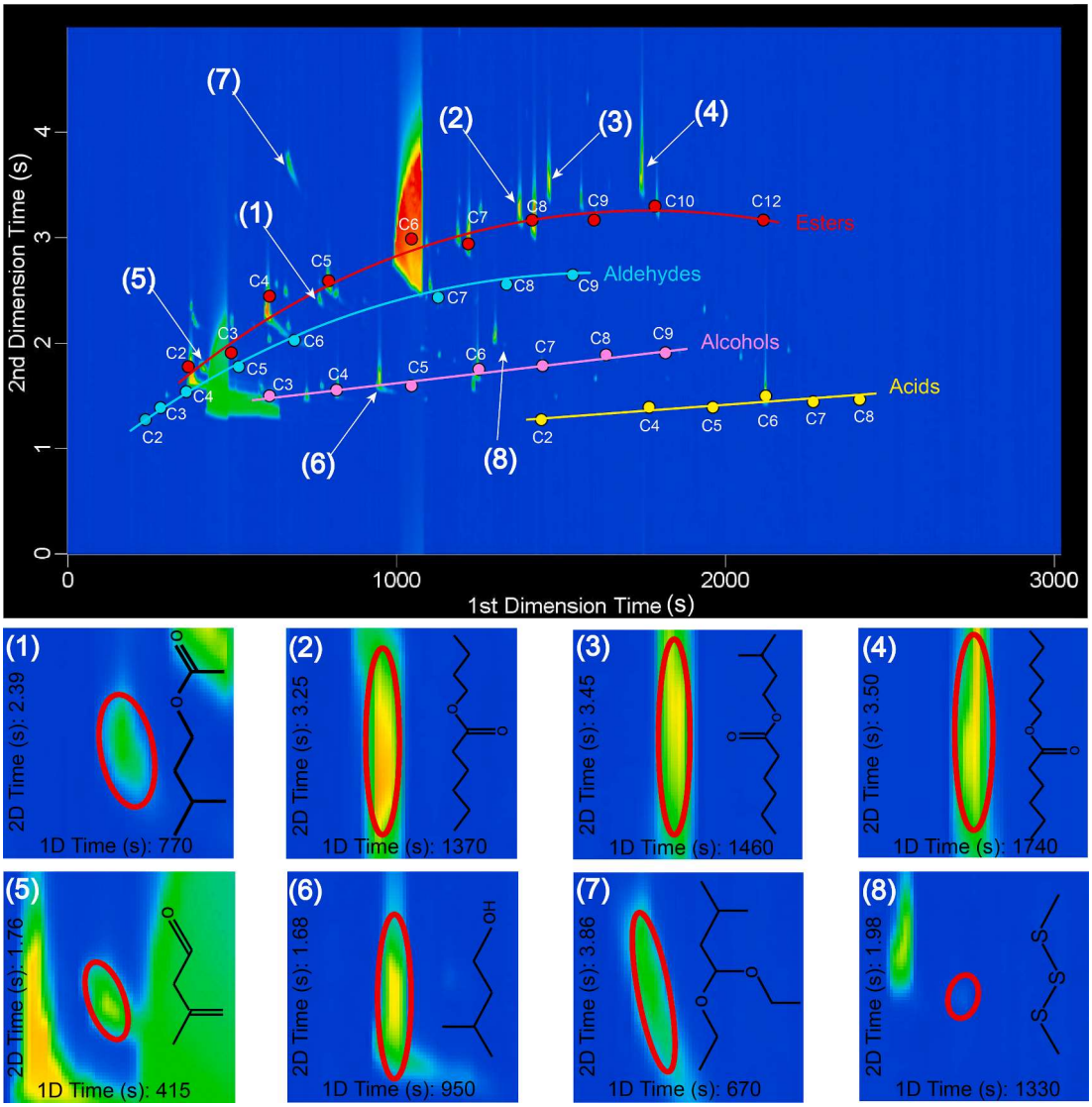


Fig. 1. Two-dimensional total ion chromatogram (TIC) and enlarged view of some volatile compounds of fresh SFB obtained from HS-SPEM-GC × GC-TOF/MS. (1) isoamyl acetate, (2) butyl hexanoate , (3) isoamyl hexanoate , (4) hexyl hexanoate , (5) 3-methyl-1-butanal, (6) 3-methyl-1-butanol, (7) 1,1-diethoxy-3-methylbutane, (8) dimethyl trisulfide.

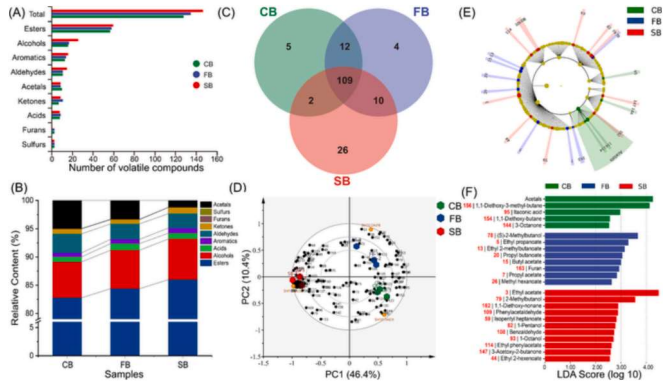


Fig. 2. Comparison of volatile compounds among the three samples based on the result of HS-SPEM-GC × GC-TOF/MS. (A) Variations in the category and number of volatile compounds. (B) Variations in the relative content of each category. (C) Variations in the diversity of volatile compounds. (D) Variations in metabolic profile based on principal component analysis. (E-F) The cladogram and characteristic volatiles of each sample obtained from LEfSe analysis. The ID of each compound was consistent with that in Table S1.

Baijiu mainly originate from the distillation of *Zaopei* while those in the latter are produced by microbiota (Du, Ji, Xing, Wang, & Xu, 2021; Ding, Wu, Huang, & Zhou, 2015). Our previous work demonstrated that bioaugmented *Daqu* could alleviate the acid stress to flavor-producing microorganisms by delaying the succession rate of *Lactobacillus* during the process, thus increasing the diversity and content of volatiles (Mu et al., 2023). However, acetals, condensates of aldehydes and alcohols, were not detected due to the limited resolution of GC–MS. Therefore, what factors mediate the enrichment of acetal in CB remains to be further studied. Additionally, 109 compounds were shared in three samples, while SB, FB and CB had 26, 4 and 5 unique compounds, respectively (Fig. 2C). The PCA bi-plot revealed that bioaugmented *Daqu* significantly changed the flavor profile of fresh SFB (Fig. 2D). The samples were distinctly separated on the PC1 (46.4%) and PC2 (10.4%), with more than half of the volatiles being gathered in the negative axis of PC1, indicating that these compounds had higher relative content in SB. This might be related to the enrichment of volatile and non-volatile metabolites in *S_Daqu* that provided abundant precursors for flavor synthesis during the process (Mu et al., 2022a).

The characteristic compounds of each sample were identified using LEfSe analysis, with 5, 8 and 11 features determined for CB, FB and SB,

respectively (Fig. 2E-F). An abundance of acetals was typical of CB, especially 1,1-diethoxy-3-methyl-butane (156) and 1,1-diethoxy-butane (154). These two compounds have been reported and the former dominated in SFB (Zheng et al., 2022), consistent with the present study (Table S1). Notably, the latter was only detected in CB, explaining the maximum of acetals observed in CB. The characteristics of FB were mainly composed of esters, including ethyl propanoate (5), propyl acetate (7), ethyl 2-methylbutanoate (13), butyl acetate (15), propyl butanoate (20) and methyl hexanoate (26). These compounds were detected in previous works, but only ethyl propanoate, ethyl 2-methylbutanoate and methyl hexanoate were identified as the odorants of SFB based on GC–O and AEDA, which presented the banana, berry and fruity odors, respectively (Zheng et al., 2022; Fan & Qian, 2005; 2006). In contrast, the features of SB were more diverse, consisting of 11 compounds in 5 classes with esters, alcohols and aromatics being the dominant ones (Fig. 1F). Ethyl acetate (3), ethyl 2-hexenoate (44) and isopentyl heptanoate (59) were commonly found in Baijiu (Zheng et al., 2022; Wang, Gao, Liu, Chen, & Xu, 2020), and the former with a fruity odor was the skeleton component of SFB and contributed to distinguish the SFB from different regions (He et al., 2020). For alcohols, the relative content of 2-methyl-1-butanol (79) was the highest, followed by 1-pentanol (82) and 1-octanol (93). They were identified as aroma compounds in Baijiu and showed alcoholic, green and fruity odors, respectively (Zhu et al., 2020; Fan & Qian, 2005; 2006). Additionally, benzaldehyde (108), phenylacetaldehyde (109) and ethyl phenylacetate (114) were also identified as characteristics of SB, which

Table 1
Aroma compounds identified by HS-SPEM-GC-O/MS in fresh SFB samples.

No.	Aroma compounds	Descriptor[a]	CAS	RI-cal[b]	RI-lit[c]	Identification[d]	FD[e]		
							CB	FB	SB
1	Ethyl acetate	pineapple	141-78-6	885	894	MS, aroma, RI, S	9	9	27
2	3-Methylbutanal	green, malty	590-86-3	919	915	MS, aroma, RI, S	27	27	27
3	Ethyl propanoate	banana, fruity	105-37-3	966	953	MS, aroma, RI	–	–	1
4	Ethyl 2-methylpropanoate	apple	97-62-1	975	961	MS, aroma, RI, S	27	27	27
5	Ethyl butanoate	pineapple	105-54-4	1038	1042	MS, aroma, RI, S	243	243	243
6	Ethyl 2-methylbutanoate	berry	7452-79-1	1053	1058	MS, aroma, RI, S	9	9	27
7	Ethyl 3-methylbutanoate	apple	108-64-5	1070	1074	MS, aroma, RI, S	27	9	9
8	1,1-Diethoxy-3-methylbutane	fruity, green	3842-03-3	1088	1068	MS, aroma, RI	–	1	–
9	2-Methyl-1-propanol	wine	78-83-1	1095	1098	MS, aroma, RI, S	3	3	1
10	Isoamyl acetate	banana	123-92-2	1125	1129	MS, aroma, RI, S	3	27	9
11	Ethyl pentanoate	fruity	539-82-2	1155	1142	MS, aroma, RI, S	81	243	81
12	1-Butanol	alcoholic	71-36-3	1171	1145	MS, aroma, RI, S	9	3	3
13	Ethyl 4-methylpentanoate	fruity	25415-67-2	1189	1197	MS, aroma, RI, S	3	27	3
14	3-Methyl-1-butanol	malt, burnt	123-51-3	1231	1225	MS, aroma, RI, S	3	9	3
15	Ethyl hexanoate	fruity, sweet	123-66-0	1247	1252	MS, aroma, RI, S	243	243	243
16	Hexyl acetate	fruity, herb	142-92-7	1281	1264	MS, aroma, RI, S	3	9	9
17	Propyl hexanoate	sweet, fruity	626-77-7	1306	1328	MS, aroma, RI, S	1	1	1
18	Ethyl heptanoate	pineapple, sweet	106-30-9	1348	1357	MS, aroma, RI, S	81	243	81
19	Ethyl lactate	mushroom	687-47-8	1361	1363	MS, aroma, RI, S	27	27	27
20	1-Hexanol	resin, green	111-27-3	1368	1366	MS, aroma, RI, S	9	27	27
21	Dimethyl trisulfide	sulfur, gasoline	3658-80-8	1375	1383	MS, aroma, RI, S	27	27	81
22	Butyl hexanoate	fruity	626-82-4	1404	1419	MS, aroma, RI, S	3	27	9
23	Ethyl octanoate	fruity, grape	106-32-1	1425	1442	MS, aroma, RI, S	243	243	243
24	Acetic acid	vinegar	64-19-7	1441	1448	MS, aroma, RI, S	9	9	9
25	Isopentyl hexanoate	sweet, cheesy	2198-61-0	1457	1460	MS, aroma, RI, S	9	27	3
26	Pentyl hexanoate	fruity	540-07-8	1493	–	MS, aroma, S	3	9	1
27	Benzaldehyde	almond	100-52-7	1505	1521	MS, aroma, RI, S	1	3	3
28	Ethyl nonanoate	fruity	123-29-5	1518	1541	MS, aroma, RI, S	1	3	3
29	Ethyl 2-hydroxy-4-methylvalerate	bedbug-like	10348-47-7	1526	1520	MS, aroma, RI, S	3	9	9
30	1-Octanol	green	111-87-5	1543	1559	MS, aroma, RI, S	1	1	1
31	Hexyl hexanoate	green, fruity	6378-65-0	1613	1615	MS, aroma, RI, S	9	27	9
32	Butanoic acid	cheesy, sour	107-92-6	1628	1623	MS, aroma, RI, S	27	81	27
33	Ethyl decanoate	fresh, fruity	110-38-3	1636	1642	MS, aroma, RI, S	9	27	27
34	Isoamyl octanoate	floral	2035-99-6	1653	–	MS, aroma, S	3	3	3
35	Ethyl *trans*-4-decenoate	earthy	76649-16-6	1660	–	MS, aroma, S	–	1	1
36	Ethyl benzoate	fruity	93-89-0	1663	1640	MS, aroma, RI, S	1	1	1
37	Diethyl butanedioate	wine, fruity	123-25-1	1670	1679	MS, aroma, RI, S	1	3	3
38	Heptyl hexanoate	bedbug-like	6976-72-3	1697	1683	MS, aroma, RI	9	1	1
39	(2,2-Diethoxyethyl) benzene	floral	6314-97-2	1708	1724	MS, aroma, RI	9	1	3
40	Ethyl phenylacetate	rosy, honey	101-97-3	1794	1793	MS, aroma, RI, S	3	27	27
41	Hexyl octanoate	fruity	1117-55-1	1816	–	MS, aroma, S	1	1	1
42	1-Azido-2-methylbenzene	sweet	31656-92-5	1824	–	MS, aroma	–	1	1
43	Hexanoic acid	sweaty, cheesy	142-62-1	1849	1845	MS, aroma, RI, S	27	81	27
44	Unknown	steamed grain	–	1863	–	aroma	3	9	3
45	Furfuryl hexanoate	green	39252-02-3	1871	1857	MS, aroma, RI	1	1	1
46	Ethyl 3-phenylpropanoate	floral	2021-28-5	1888	1897	MS, aroma, RI, S	3	27	9
47	Phenethyl alcohol	rosy, honey, lilac	60-12-8	1908	1910	MS, aroma, RI, S	1	1	1
48	4-Methylphenol	mud-like, animal	106-44-5	2045	2083	MS, aroma, RI, S	1	9	9
49	Ethyl hexadecanoate	waxy, sweet	628-97-7	2242	2264	MS, aroma, RI, S	9	9	9

[a] Odor quality perceived at the sniffing port.
[b] Calculated retention index (RI) of aroma compounds on DB-WAX based on C8-C40.
[c] Reported RI of aroma compounds on DB-WAX based on literatures.
[d] Aroma compounds were identified based on RI, NIST standard mass spectrometry (MS), aroma (aroma description compared to standard chemicals) and standard chemicals (S).
[e] Flavor dilution factor (FD).

are known to impart floral, fruity and honey odors to Baijiu (Du, Li et al., 2021; Zhao et al., 2018). More importantly, phenylacetaldehyde and ethyl phenylacetate were found to have a significant impact on the aftertaste of SFB during consumption (Chen et al., 2023). These results suggested that the volatile profiles of fresh SFB can be enhanced by using bioaugmented *Daqu* under identical brewing conditions.

3.2. Effects of Daqu attriubutes on aroma compounds of fresh SFB

As shown in Table 1, 49 aroma compounds were identified by HS-SPME-GC–O/MS, including 29 esters, 7 aromatics, 5 alcohols, 3 acids, 1 aldehyde, 1 acetal, 1 phenol, 1 sulfur-containing compound and 1 unknown compound. The number of aroma compounds varied by *Daqu* attributes, with 45, 48 and 47 compounds detected in CB, FB and SB, respectively. Ethyl *trans*-4-decenoate and 1-azido-2-methylbenzene were not detected in CB, while ethyl propanoate and 1,1-diethoxy-3-methylbutane were only found in SB and FB, respectively. The intensity of these aroma compounds was further evaluated using AEDA, and the variations in FD value among samples are shown in Table 1.

Esters were the most important aroma compounds in all samples, and ethyl hexanoate, ethyl butanoate and ethyl octanoate with the highest FD value (243) have been reported as key contributors to the fruity and sweet odors of SFB, especially the former (Zhao et al., 2018; Fan & Qian, 2006). Ethyl hexanoate was also a critical odorant in Jian- and sesame-flavor Baijiu (Du, Li et al., 2021; Li et al., 2019), but its FD value was low in light-flavor Baijiu (Wang, Ming et al., 2021). The highest FD values of ethyl pentanoate (243) and ethyl heptanoate (243) were observed in FB, which are similar to that of *Laobaigan* (243 and 243) and *Jiannanchun* (256 and 512) Baijiu (Zhu et al., 2020; Fan & Qian, 2006). Meanwhile, the FD values of eight esters with fruity and sweet odors were also the highest in FB, including ethyl 4-methylpentanoate (27), butyl hexanoate (27), and isopentyl hexanoate (27), hexyl hexanoate (27) and pentyl hexanoate (9). The highest FD values of ethyl acetate (27) and ethyl 2-methylbutanoate (27) in SB enhanced its berry and apple aroma. By contrast, CB had the highest FD values of ethyl 3-methylbutanoate (27) and heptyl hexanoate (9), resulting in a fruity aroma with a slightly bedbug-like odor. In addition, the FD values of five esters, namely hexyl acetate, ethyl nonanoate, ethyl 2-hydroxy-4-methylvalerate, ethyl decanoate and diethyl butanedioate, were higher in FB and SB than that in CB.

The highest FD value among the five identified alcohols was found for 1-hexanol, with FD values of 9, 27 and 27 in CB, FB and SB, respectively. The highest FD value of 3-methyl-1-butanol was observed in FB, while that of 1-butanol was in CB. For acids, the FD values of hexanoic acid and butanoic acid with cheese and sour odors in FB were higher than those in CB. As we all know, alcohols are produced by the metabolism of sugars and amino acids during fermentation (Hazelwood et al., 2008), whereas acids are primarily derived from the metabolism of bacteria and yeast, as well as the oxidation of ethanol and acetaldehyde (Chen et al., 2021). The biotic esterification of these two components during *Zaopei* fermentation is the primary pathway to increase the concentration of esters in Baijiu (Kruis et al., 2019). Our previous work indicated that F_*Daqu* and S_*Daqu* could promote the synthesis and accumulation of volatile compounds such as esters, alcohols and acids in *Zaopei* (Mu et al., 2023). Therefore, the differences in FD values for these aroma compounds among the samples were closely related to the *Daqu* attributes.

The FD values of aromatics such as ethyl phenylacetate, ethyl 3-phenylpropanoate and benzaldehyde were higher in both FB and SB, contributing sweet, floral and almond odors, respectively. However, 2,2-diethoxyethyl-benzene with an FD value of 9 was the most important aromatic in CB. Besides, dimethyl trisulfide had a notably high FD value of 81 in SB, and the FD value of 4-methylphenol, a key volatile component of pit mud, was also higher in SB (9) and FB (9) than in CB (1). The content of the former was higher in the artificial pit mud cultured with bioaugmented *Daqu* (Mu et al., 2022b), and the latter was usually abundant in the aged pit mud (>30 years) (Zhang et al., 2020). These two compounds are known to contribute significantly to the flavor feature of high-quality fresh SFB from aged mud pits and high-grade finished products (Dong et al., 2019; Zhao et al., 2018), supporting the results that bioaugmented *Daqu* drive the evolution of community and metabolism in brewing microecology (Chen et al., 2020).

3.3. Quantitative characteristion of aroma compounds

Differences in the contents of aroma compounds among the samples were identified using DI-GC–FID and DI-GC–MS/MS (Table S2). Similar to previous works (Hong et al., 2021; Dong et al., 2019), eight skeleton compounds, including ethyl hexanoate, ethyl acetate, ethyl lactate, acetic acid, ethyl butanoate, hexanoic acid, 3-methyl-1-butanol and 1-butanol, exhibited an average content of over 100 mg/L. However, there were significant differences in their contents among the samples, especially in ethyl hexanoate and ethyl acetate, the characteristic of SFB (Fig. 3A). Specifically, FB exhibited the highest contents of ethyl hexanoate, ethyl butanoate, hexanoic acid and 3-methyl-1-butanol, while SB had the highest contents of ethyl acetate and acetic acid. Besides, the contents of ethyl hexanoate and ethyl butanoate in SB were also considerably higher than that in CB, endowing FB and SB with stronger fruity and sweet odors (Zhao et al., 2018). In contrast, the highest levels of ethyl lactate and 1-butanol were found in CB, which can contribute to over-ripe fruity and mellow aromas in Baijiu, but excessive amounts will cause bitterness and astringency (Hong et al., 2021; Jia et al., 2020). Therefore, the ratio of ethyl hexanoate to ethyl lactate is a key indicator of SFB quality, with values of 2.46, 3.81 and 3.59 in CB, FB and SB, respectively. Additionally, the content of ethyl hexanoate in FB (6680 mg/L) and SB (6130 mg/L) was similar to that of the fresh SFB produced from the aged mud pit (Dong et al., 2019), indicating that their aroma quality had been improved. These two ethyl esters are formed by the esterification of corresponding organic acids with ethanol. Gao et al. (2021) confirmed a synergistic effect of ethyl hexanoate formation between esterases in *Daqu* and microbiota in pit mud, with S_*Daqu* and F_*Daqu* exhibiting significantly improved esterifying ability (Mu et al., 2022a). Furthermore, our previous study showed that these two bioaugmented *Daqu* could reduce the relative abundance of *Lactobacillus* in *Zaopei* through changes in community succession rate, leading to an increase in ethyl hexanoate and a decrease in ethyl lactate (Mu et al., 2023), which was also observed in FB and SB (Fig. 3A).

As shown in Fig. 3B, six of the nine compounds with an average content of 10 to 100 mg/L, including butanoic acid and ethyl pentanoate with cheesy and fruity odors, had the highest levels in FB, while 1-hexanol with the green odor was the most abundant in SB. Besides, butanoic acid exceeded 100 mg/L only in FB, while ethyl 2-hydroxy-4-methylvalerate and butyl hexanoate in CB were <10 mg/L (Table S2). Butanoic acid is a key short-chain fatty acid produced by functional microorganisms in pit mud, which can be esterified with ethanol to form ethyl butanoate, or transformed into hexanoate acid through chain elongation and then further into ethyl hexanoate (Qian et al., 2021). Compounds with an average content of 1 to 10 mg/L were dominated by hexyl hexanoate, 3-methylbutanal, ethyl 2-methylpropanoate, isopentyl hexanoate, hexyl acetate and ethyl 3-methylbutanoate, with most of them having the highest levels in FB, while the highest content of 3-methylbutanal and ethyl 3-methylbutanoate was observed in SB and CB, respectively (Fig. 3C). Among them, hexyl hexanoate and ethyl 3-methylbutanoate were identified as the major contributors to the sweetness of Baijiu (Sun, Ma, Chen, Xu, & Tang, 2021). In contrast, 3-methylbutanal was one of the main contributors to Baijiu pungency (He, Tang, Yu, Chen, & Xu, 2022), and its high content in SB might be related to the significant enrichment of leucine in S_*Daqu* (Mu et al., 2022a; Hazelwood et al., 2008). The distribution of aroma compounds with low content (<1 mg/L) also differed across the samples (Fig. 3D). For example, the contents of six compounds, such as 1-octanol, dimethyl trisulfide and 4-methylphenol, were higher in SB, which had the aroma

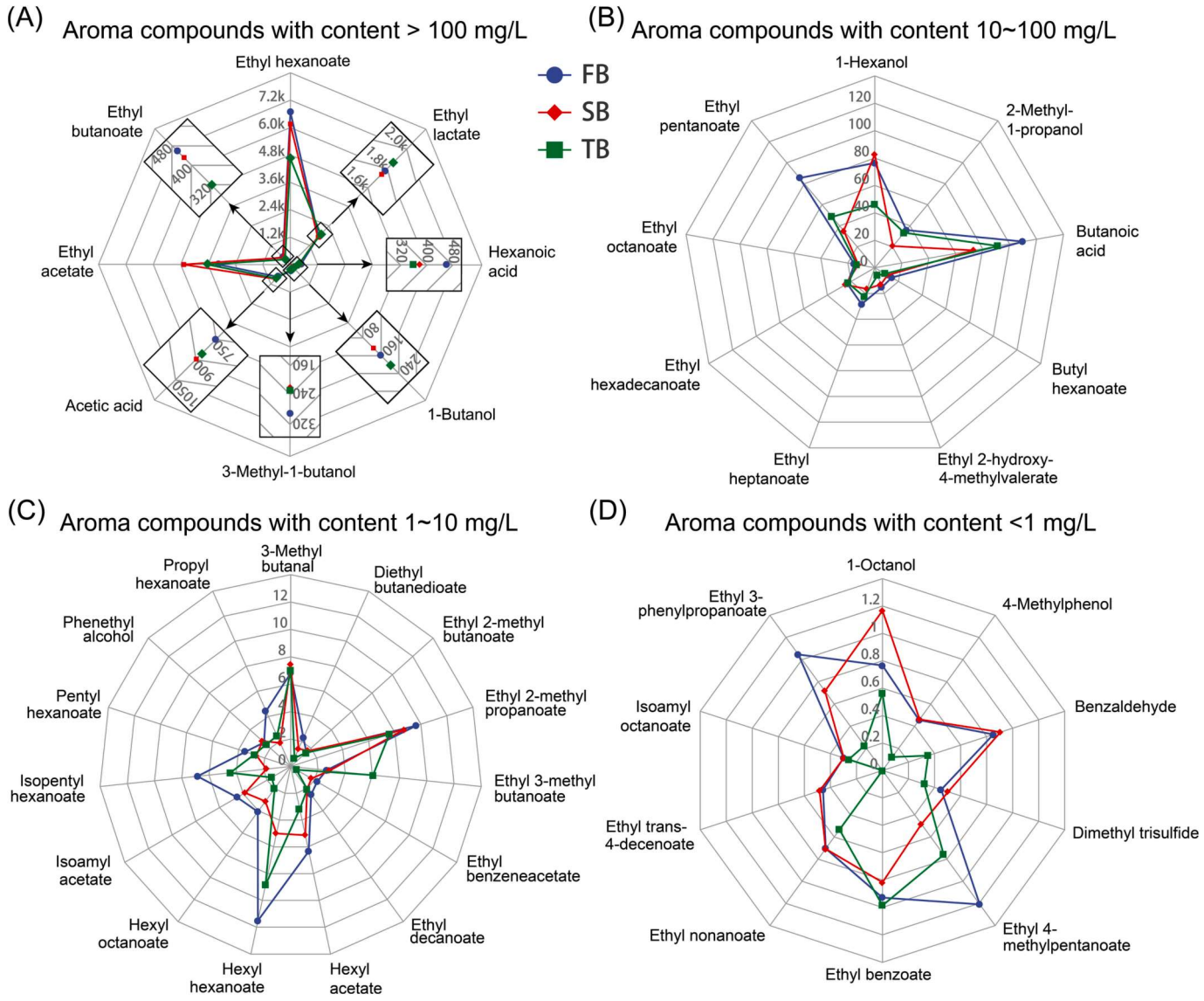


Fig. 3. Differences in the content of aroma compounds in the three fresh SFB samples. (A) Eight compounds with an average content of >100 mg/L. (B) Nine compounds with an average content between 10 and 100 mg/L. (C) Fifteen compounds with an average content between 1 and 10 mg/L. (D) Ten compounds with an average content of <1 mg/L.

characteristics of green, gasoline and mud-like, respectively. Conversely, the contents of three compounds, namely ethyl 4-methylpentanoate, ethyl 3-phenylpropanoate and isoamyl octanoate, were higher in FB, while that of ethyl benzoate was higher in CB. Aliphatic alcohols are typically formed through lipid oxidation and their content was significantly positively correlated with *Bacillus* abundance (Seo et al., 2018). Therefore, the highest levels of 1-octanol and 1-hexanol in SB and the lowest levels in CB could be attributed to the differences in the relative abundance of *Bacillus* in the corresponding *Daqu* (Mu et al., 2022a). These results indicated that the main characteristic aromas of FB were ethyl hexanoate, butanoic acid, ethyl pentanoate, hexyl hexanoate, isopentyl hexanoate, ethyl 4-methylpentanoate and ethyl 3-phenylpropanoate, those of SB were ethyl hexanoate, ethyl acetate, 1-hexanol and 1-octanol, while those of CB were ethyl lactate, 1-butanol and ethyl 3-methylbutanoate.

3.4. Determination of aroma-active compounds based on OAVs

The OAVs were calculated to explore the effect of the differential compounds mentioned above on the sensory properties and 29 aroma-active compounds were determined based on OAV ≥ 1, including 17 esters, 4 alcohols, 3 acids, 2 aromatics, 1 aldehyde, 1 phenol and 1 sulfide-containing compound (Table 2). Among them, the OAV of ethyl hexanoate was the highest in all samples (84,503 ~ 121,499), followed by ethyl butanoate (3,890 ~ 5,580) and ethyl pentanoate (1,306 ~ 3,162). This result was consistent with Dong et al. (2019), indicating that these compounds significantly affected the aroma features of SFB. Notably, although the content of dimethyl trisulfide was <1 mg/L in all samples (CB 0.321 mg/L; FB 0.455 mg/L; SB 0.461 mg/L), its OAV ranged from 892 to 1,401 due to an extremely low odor threshold (0.36 μg/L). Dimethyl trisulfide was mainly derived from the degradation of sulfur-containing amino acid and also detected in other SFB (*Yanghe Daqu* and *Jiannanchun*), but showed a low FD value and weak contribution to flavor (Fan & Qian, 2005; 2006). However, it has been determined as one of the characteristics of sauce- and sesame-flavor Baijiu (Ma et al., 2020; Li et al., 2019) and endowed a unique aromatic flavor called Chen-aroma in light-flavor Baijiu (Wang, Ming et al., 2021). The present results showed that the contribution of dimethyl trisulfide to the aroma profile of the fresh Baijiu was only followed by ethyl hexanoate, ethyl butanoate and ethyl pentanoate, and the

Table 2
Odor activity values (OAVs) of aroma-active compounds in fresh SFB samples.

No.[a]	Aroma compounds	Odor thresholds (μg/L)[b–f]	OAVs			
			CB	FB	SB	Average
15	Ethyl hexanoate	55[b]	84,503	121,499	111,532	105,845
5	Ethyl butanoate	82[b]	3,890	5,580	5,247	4,906
11	Ethyl pentanoate	27[b]	1,792	3,162	1,306	2,087
21	Dimethyl trisulfide	0.36[c]	892	1,254	1,401	1,182
23	Ethyl octanoate	13[b]	1,022	1,158	937	1,039
7	Ethyl 3-methylbutanoate	7[b]	864	376	408	549
2	3-Methyl-1-butanal	16.5[e]	423	412	452	429
43	Hexanoic acid	2520[c]	143	183	150	158
4	Ethyl 2-methylpropanoate	57.5[e]	131	167	150	149
1	Ethyl acetate	32600[b]	112	97	143	117
32	Butanoic acid	964[b]	95	114	76	95
6	Ethyl 2-methylbutanoate	18[b]	83	86	94	88
12	1-Butanol	2730[b]	74	58	45	59
10	Isoamyl acetate	94[c]	17	48	41	35
20	1-Hexanol	5370[b]	9	14	15	13
19	Ethyl lactate	128000[b]	15	14	13	14
46	Ethyl 3-phenylpropanoate	125[b]	2	8	6	5
24	Acetic acid	160000[b]	5	5	6	5
31	Hexyl hexanoate	1890[b]	5	6	3	4
40	Ethyl phenylacetate	407[b]	1	5	4	4
25	Isopentyl hexanoate	1400[f]	3	5	1	3
16	Hexyl acetate	1500[d]	2	4	3	3
13	Ethyl 4-methylpentanoate	389[d]	2	3	1	2
22	Butyl hexanoate	5250.31[d]	2	3	2	2
48	4-Methylphenol	167[f]	<1	3	3	2
33	Ethyl decanoate	1120[d]	2	2	2	2
18	Ethyl heptanoate	13200[f]	2	2	1	2
14	3-Methyl-1-butanol	179000[d]	1	2	1	1
30	1-Octanol	1100[e]	<1	<1	1	<1
9	2-Methyl-1-propanol	40000[c]	<1	<1	<1	<1
36	Ethyl benzoate	1400[e]	<1	<1	<1	<1
17	Propyl hexanoate	12800[b]	<1	<1	<1	<1
29	Ethyl 2-hydroxy-4-methylvalerate	45953.66[d]	<1	<1	<1	<1
28	Ethyl nonanoate	3200[e]	<1	<1	<1	<1
27	Benzaldehyde	4200[e]	<1	<1	<1	<1
47	Phenethyl alcohol	28900[e]	<1	<1	<1	<1
37	Diethyl butanedioate	353000[f]	<1	<1	<1	<1
49	Ethyl hexadecanoate	—	—	—	—	—

[a] The No. of aroma compounds are the same as in Table 1.
[b~f] Odour thresholds were taken from Li et al. (2019), Wang, Ming et al. (2021), Du, Li et al. (2021), Chen et al. (2021) and Zhao et al. (2018), respectively.

contribution to both FB and SB was evidently higher than CB. Similarly, 3-methylbutanal with green and malty odors ranked at seven places due to the higher OAVs (412 ~ 452), but it has not been reported as an aroma-active compound in SFB, and its FD value in some famous SFB was only 8 ~ 16 (Hong et al., 2021; Zhao et al., 2018; Fan & Qian, 2005; 2006). Recently, 3-methylbutanal was also determined as one of the most critical odorants in *Xiaoqu* Baijiu, with an OAV range of 190 to 307 (Chen et al., 2021), indicating that it might play an important role in shaping the aroma feature of fresh SFB. Hexanoic acid and butanoic acid were the aroma-active compounds and key precursors in SFB, with a much higher OAV than acetic acid, consistent with previous works (Hong et al., 2021; Dong et al., 2019). Meanwhile, hexanoic acid was more difficult to release into the oral cavity and contributed more to the aftertaste during Baijiu drinking (Chen et al., 2023). Additionally, 1-octanol was an essential aroma compound with a rosy odor in the SFB aroma profile (Zhao et al., 2018), but its OAV exceeded one only in SB. On the other hand, the OAV of 4-methylphenol was three in both SB and FB but was less than one in CB, supporting the results of AEDA. However, the contribution of ethyl heptanoate and ethyl 3-methylbutanoate evaluated by AEDA and OAV was contradictory, possibly due to the composition of matrices and interactions between aroma compounds (Lemarcq et al, 2021).

3.5. Identification of the discriminable markers among samples

The PCA model was performed to compare the differences among fresh SFB samples, and the results suggested that CB, FB and SB were clearly separated from the score plot, with PC1 and PC2 explaining the total variance of 53.6% and 38.1%, respectively (Fig. S1A). Hence, the OPLS-DA model was used to further identify the characteristics of each pair of samples. As shown in Fig. 4A-C, the superior models had good fitting and predictability, as indicated by R2 and Q2 values above 0.9 in each model. Moreover, 200 permutation tests showed that the models were not overfitted and could be used to screen discriminable markers between different samples (Fig. S1B-D). Therefore, the differential compounds were identified by variable importance in projection (VIP) score (>1.0) and statistical significance (ANOVA: $p < 0.05$). Among the 17 differential compounds between CB and FB, ethyl 3-methylbutanoate was the marker of the former, while the other 16 compounds were significantly up-regulated in the latter, including skeleton components such as ethyl hexanoate, ethyl butanoate and ethyl pentanoate (Fig. 4D). However, compounds with higher content and OAV exhibited lower VIP score, such as ethyl hexanoate and ethyl butanoate with VIP scores of 1.106 and 1.104, respectively. In contrast, ethyl 2-hydroxy-4-methylvalerate, with lower content (5.94 ~ 15.2 mg/L) and OAV (<1), showed the highest VIP score of 1.697 (Table S3). Similarly, among the differential compounds identified between CB and SB, VIP scores of ethyl hexanoate, ethyl butanoate, ethyl pentanoate and ethyl 3-methylbutanoate ranged from 1.023 to 1.128, while ethyl 2-hydroxy-4-methylvalerate had the highest score of 1.622 (Table S3). Ethyl 2-hydroxy-4-methylvalerate, a common aroma compound, had lower OAV in various Baijiu due to a higher threshold (45953.66 μg/L) (Dong et al., 2019; Du, Li et al., 2021).

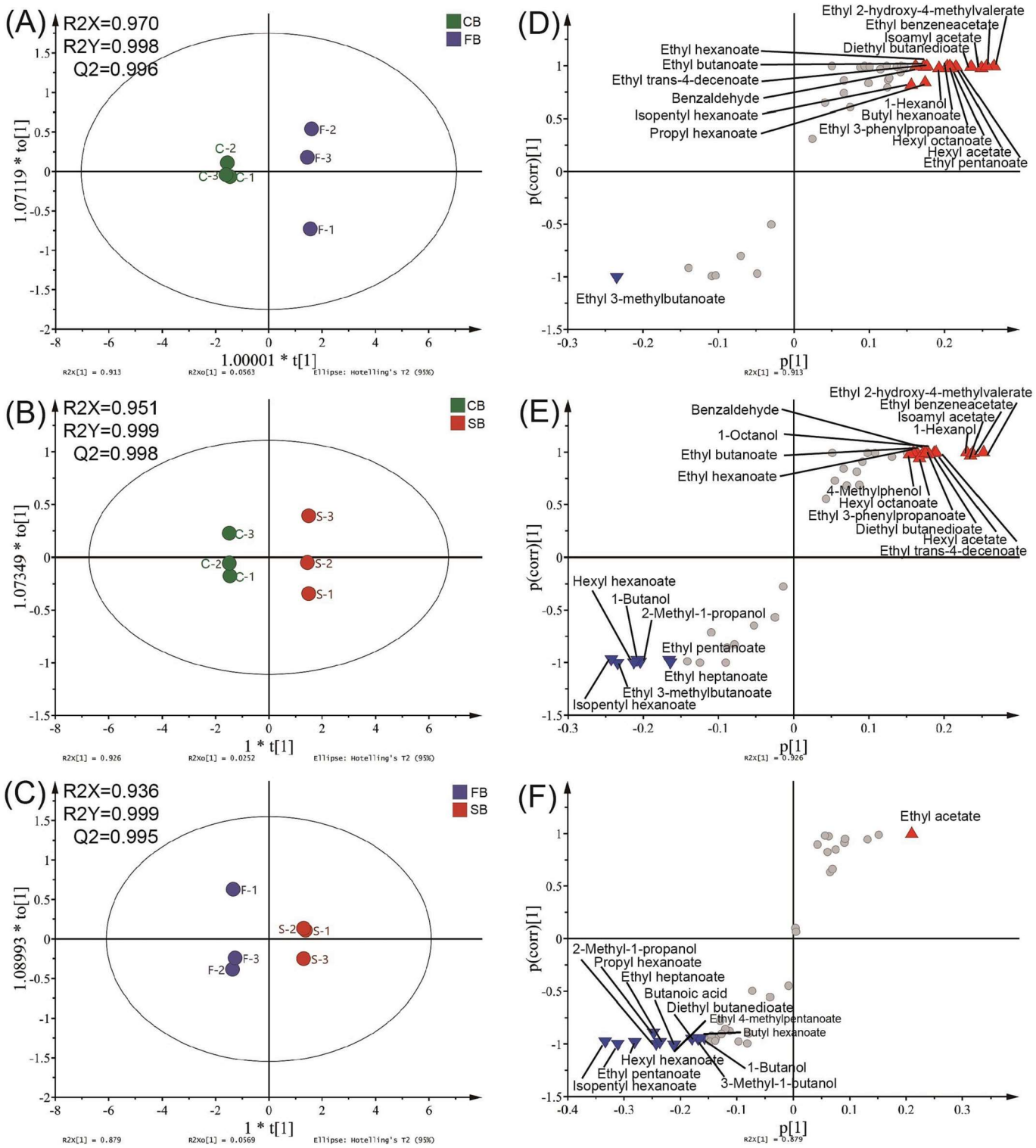


Fig. 4. The score plot (A-C) and S-plot (D-F) of OPLS-DA based on the quantitative data set. Significantly different metabolites were labeled in the S-plot as blue for down-regulated, red for up-regulated, and gray for no significant change. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Recently, it was also determined as the significant differential compound between sakes with different brewing processes, despite that its concentration was only 6.80 ~ 39.0 μg/L (Sasaki, Ochiai, Yamazakia, & Sasamoto, 2023). Therefore, compounds that contribute less to aroma should also be highly valued in sample discrimination.

Additionally, 1-butanol and 2-methyl-1-propanol were characteristics of CB, while 4-methylphenol, a metabolite of *Clostridium butyricum* contributing to the mud-like odor of SFB (Du et al., 2017; Zhao et al., 2018), was a feature of SB (Fig. 4E). Similarly, three main higher alcohols in Baijiu, namely 1-butanol, 3-methyl-1-butanol and 2-methyl-1-propanol, were also up-regulated in FB compared with SB (Fig. 4F). Excessive higher alcohols significantly affect the comfort and safety of Baijiu, and 1-butanol and 3-methyl-1-butanol as aroma-active compounds only contributed to the burst-stage of Baijiu perception (Chen

et al., 2023). Prior works have highlighted that the bioturbation technology based on the interspecies interaction effectively reduced the content of higher alcohols in Baijiu (Zhu et al., 2022), which was supported by the present results. Besides, isopentyl hexanoate with the highest VIP score of 2.143 was the feature of FB, while ethyl acetate was the only compound with VIP > 1 in SB (Fig. 4F). These results suggested that the differences in the content of ethyl 3-methylbutanoate and ethyl 2-hydroxy-4-methylvalerate could be used to discriminate CB from FB and SB, and that of isopentyl hexanoate and ethyl acetate were the key to distinguish the latter two.

4. Conclusion

In this study, GC × GC-TOF/MS and GC–O/MS were used to identify 168 volatile compounds and 49 aroma compounds, whose diversity and content were significantly influenced by bioaugmented *Daqu*. Among the 29 aroma-active compounds identified, the top five were ethyl hexanoate, ethyl butanoate, ethyl pentanoate, dimethyl trisulfide and ethyl octanoate. Notably, bioaugmented *Daqu* increased the content of ethyl hexanoate, ethyl butanoate, dimethyl trisulfide and 4-methylphenol, which are associated with fruity, pineapple, mud-like and Chen odors in FB and SB, indicating the potential of bioaugmentation strategy in improving the flavor quality of Baijiu. Multivariate statistical analysis indicated that the discriminators among the three fresh SFB samples were isopentyl hexanoate, ethyl acetate and ethyl 2-hydroxy-4-methylvalerate. These findings provide a theoretical basis for characterizing and optimizing Baijiu production.

CRediT authorship contribution statement

Yu Mu: Data curation, Software, Methodology, Writing – original draft. **Jun Huang:** Supervision, Resources. **Rongqing Zhou:** Conceptualization, Resources, Formal analysis, Supervision, Writing – review & editing. **Suyi Zhang:** Conceptualization, Supervision. **Hui Qin:** Project administration, Formal analysis. **Hanlan Tang:** Investigation, Data curation. **Qianglin Pan:** Investigation, Software. **Huifang Tang:** Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research was supported by the Cooperation Project of Luzhou Lao Jiao Co., Ltd. and Sichuan University (No. 21H0997).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodchem.2023.136429.

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